

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO4: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 30%

Dr G A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Mock Interview

Duration: 1hr

1. A machine learning engineer applies Reinforcement Learning to solve sequential decision-making problems. Explain how an agent learns optimal behavior through interaction with the environment. **(5 Marks)**
2. If you are asked to design a Reinforcement Learning solution for a real-time system such as ride-sharing or delivery optimization, explain how you would formulate the problem using a Markov Decision Process. **(5 Marks)**
3. Compare model-free and model-based Reinforcement Learning approaches. Explain their working principles and how they handle uncertainty. **(5 Marks)**
4. A recommendation system using Reinforcement Learning is underperforming. Explain how reward sparsity and hyperparameter tuning can be diagnosed and improved. **(5 Marks)**

5. If you are implementing an RL-based game-playing AI, describe how you would design the learning pipeline and ensure training stability. **(5 Marks)**
6. A company wants to use Reinforcement Learning for dynamic pricing. Explain how the agent learns from interaction and improves pricing decisions over time. **(5 Marks)**
7. If an RL model is not converging properly, explain how you would identify and fix issues related to learning rate and exploration strategy. **(5 Marks)**
8. A robotics system uses Reinforcement Learning for navigation. Explain how the environment and agent interaction leads to optimal path learning. **(5 Marks)**
9. If you are asked to improve an RL model with delayed rewards, explain techniques to handle credit assignment problems. **(5 Marks)**
10. A financial firm uses Reinforcement Learning for trading decisions. Explain how uncertainty and risk can be handled in the learning process. **(5 Marks)**

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately